UNIT I

Introduction, Failures, LEFM / EPFM & Fatigue Crack Growth

EFM course outline | spectacular failures | LEFM vs EPFM | fatigue crack growth model

I. EFM Course Outline

Fracture mechanics is the study of how cracks / flaws in a structure grow and when they cause **fracture**. It gives a quantitative link between applied stress, flaw size and material toughness.

Unit	Main topics
I	Failures, LEFM & EPFM intro, fatigue crack growth model
II	Crack growth & fracture mechanisms, Griffith theory, energy release rate
III	Elasticity review, Westergaard solution (Mode I), K-G relation
IV	Multi-parameter stress field, mixed modes, SIFs, Irwin & Dugdale models
V	Toughness testing, Paris law, closure, J-integral, FAD, mixed mode, arrest & repair

Why study Fracture Mechanics ?

> Conventional design assumes a **flaw-free** material: design stress = σ_y / FoS. This ignores cracks.

> Real structures always contain flaws (welds, inclusions, machining marks, corrosion pits).

> Fracture can occur at stresses **far below yield** (e.g. Liberty ships, Comet).

> FM answers 3 questions: (1) residual strength with a crack, (2) critical crack size, (3) how long a crack takes to grow from a_0 to a_c (inspection interval).

I.1 Three variables of fracture design

Conventional approach has ONE variable (stress σ vs strength σ_y). The fracture mechanics approach has THREE:

Applied stress σ , Flaw size a , Fracture toughness K_c

- Fracture occurs when $K = \sigma Y \sqrt{(\pi a)} \geq K_c$ (K_c = material property, units MPa$\sqrt{m}$).
- Given any two variables, the third (critical) value can be found.
- **Fracture toughness K_c** = ability of a material containing a crack to resist fracture.

2. Spectacular Failures (Case Studies)

Learn these as: **what happened → cause → lesson**. Very common 5/10 mark question.

Failure / year	Cause (fracture mechanics view)	Lesson
Liberty ships (USA, 1943-47) ~2700 welded ships	Brittle fracture: welded hull (cracks cross plates), notch-brittle steel below DBTT in cold seas, stress concentration at square hatch corners. ~1500 serious fractures, about a dozen ships lost; some broke in two while docked.	Material toughness & DBTT; crack arresters; rounded hatch corners; Charpy test specified
de Havilland Comet I (1953-54)	Two aircraft broke up in flight. **Fatigue cracks** started at corners of square ADF windows (high Kt) under repeated cabin pressurisation; riveted-hole stress concentration.	Fatigue + stress concentration; full-scale fatigue testing; rounded windows; fail-safe design
Silver Bridge, Ohio (1967)	Eyebar chain failed: a tiny crack in one eyebar grew by **stress-corrosion / corrosion fatigue** to critical size. 46 deaths.	Non-redundant (single load path) structure; inspection
F-111 (USAF, 1969)	Wing failed at well below design load from a pre-existing forging flaw in the wing pivot fitting (D6AC steel).	Led to **damage tolerance** requirement (assume initial flaws)
Alexander Kielland rig (Norway, 1980)	Fatigue crack from a fillet weld of a hydrophone support in a brace; brace failed, platform capsized. 123 deaths.	Weld fatigue, poor inspection, no redundancy
Aloha Airlines 243 (1988)	Fuselage upper section tore off: **multiple-site damage** (MSD) of lap-joint rivet holes, corrosion & disbonding in an ageing aircraft.	Ageing aircraft programme; MSD, widespread fatigue damage
Eschede train (Germany, 1998)	Fatigue crack in a wheel tyre of a high-speed train; 101 deaths.	Inspection & fatigue crack growth analysis
Titanic (1912)	Brittle steel (high S, P) + cold water below DBTT; rivet failures (slag-rich iron rivets) contributed.	Ductile-brittle transition & material choice

Common causes seen in all failures

> Pre-existing flaw or stress concentrator (sharp notch, weld defect, rivet hole).
> Low-toughness material or low service temperature (below DBTT).
> Cyclic loading (fatigue), corrosive environment (SCC, hydrogen embrittlement).
> Lack of redundancy and of regular NDT inspection.

2.1 Evolution of design philosophy

Philosophy	Idea	Example
Safe-life	Component retired after N_f / scatter factor; no cracks allowed	Landing gear, helicopter rotors
Fail-safe	Structure has redundant paths; failure of one member is carried by others	Aircraft wing, multi-spar
Damage tolerant	Assume an initial flaw a_0; predict growth $a_0 \rightarrow a_c$; inspect before a_c is reached	Modern aircraft, pressure vessels, pipelines

3. Introduction to Fracture Mechanics

3.1 Why stress concentration is not enough

- For an elliptical hole, $\sigma_{max} = \sigma(1 + 2a/b)$. For a sharp crack, $b \rightarrow 0$ so $\sigma_{max} \rightarrow \infty$ (stress singularity).
- A stress-based criterion ($\sigma_{max} = \sigma_u$) would predict failure at ~zero load – not true.
- Hence we need a **single-parameter** that characterises the crack-tip field: **K** (stress intensity factor) or **G** (energy release rate). Both are discussed in Units 2-3.

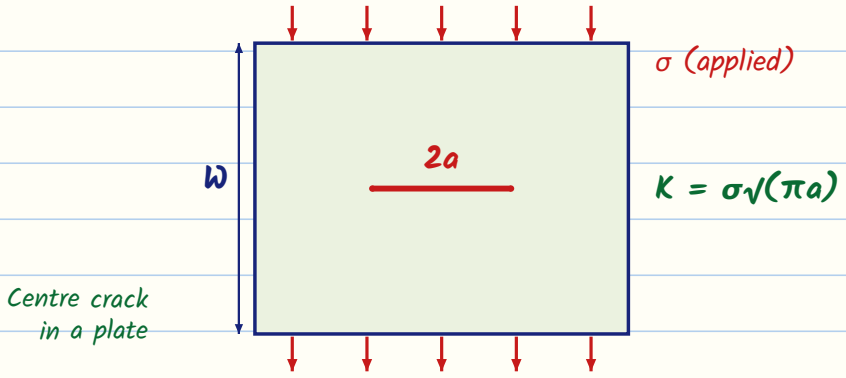


Fig 1.1 Plate with a central through crack under remote tension σ

3.2 Modes of crack loading

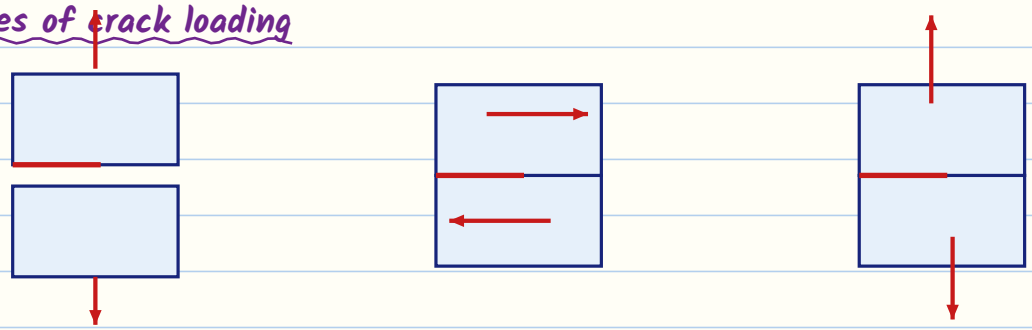


Fig 1.2 Three fundamental modes of fracture

- **Mode I** (opening): tensile stress ⊥ crack plane. Most important & most dangerous; K_I,

K_{Ic}.

- **Mode II** (sliding): in-plane shear, ⊥ to crack front. K_{II}.
- **Mode III** (tearing): out-of-plane shear, ∥ to crack front. K_{III}.
- Real cracks are often **mixed mode** (I + II + III) – superposition is applicable for linear elastic bodies.

3.3 Brittle vs ductile fracture

Brittle fracture	Ductile fracture
Little/no plastic deformation, sudden	Large plastic deformation, necking
Flat, shiny (cleavage) surface; chevron / river marks	Rough, dull, fibrous surface; cup-and-cone
Fast crack (~ 0.4 of sound speed), no warning	Slow, stable crack growth; warning given
Low energy absorbed; high strain-rate, low T, thick sections, triaxial stress	High energy absorbed; high T, thin sections

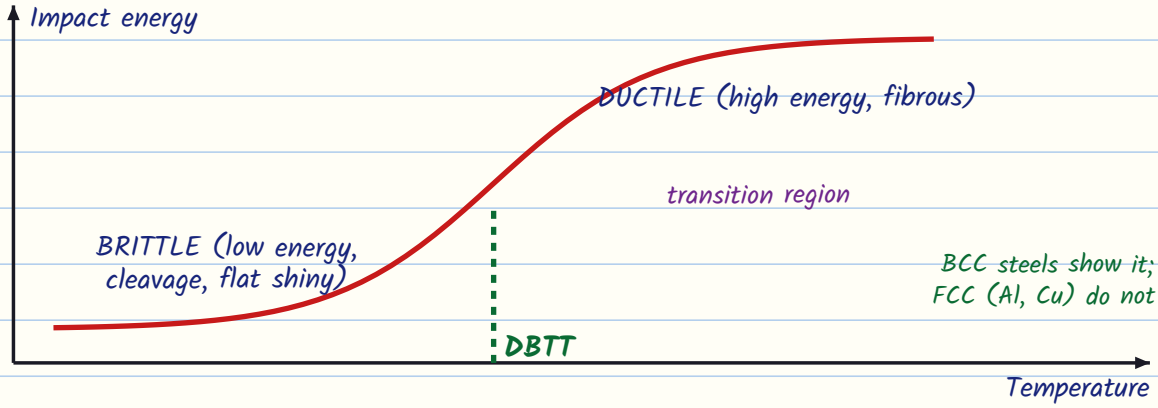


Fig 1.3 Ductile-to-brittle transition temperature (DBTT)

DBTT is found from the **Charpy V-notch impact** test (energy absorbed vs temperature). Steels in structures should operate above DBTT (e.g. ~ 27 J level).

4. LEFM and EPFM

4.1 LEFM – Linear Elastic Fracture Mechanics

- Applies when plastic zone at crack tip is **small** compared with crack length & specimen dimensions (small-scale yielding, SSY): $r_p << a, (W-a), B$.
- Whole body behaves linearly elastic; the crack-tip field is governed by **K** (or G).
- Assumptions: isotropic, homogeneous, linear elastic material; sharp crack; quasi-static loading; stress singularity ∝ $1/\sqrt{r}$.
- Used for: high-strength steels, Al/Ti alloys, ceramics, glass, thick sections (plane strain), low-temperature steel.

Criterion: fracture when $K_I = K_{Ic}$ (or $G = G_c$). $K = \sigma Y \sqrt{(\pi a)}$.

4.2 EPFM – Elastic-Plastic Fracture Mechanics

- Applies when **large-scale yielding** occurs at the tip (r_p comparable to a or ligament): ductile metals, low-strength steels, thin sections, high temperatures.
- K no longer unique. Uses **J-integral** (Rice) and **CTOD / δ** (Wells) as fracture parameters.
- Criteria: $J = J_{Ic}$, $\delta = \delta_c$. Resistance curves (J-R, δ-R) describe stable tearing.

LEFM: r_p << a, b
EPFM: r_p ~ a (use J, CTOD)

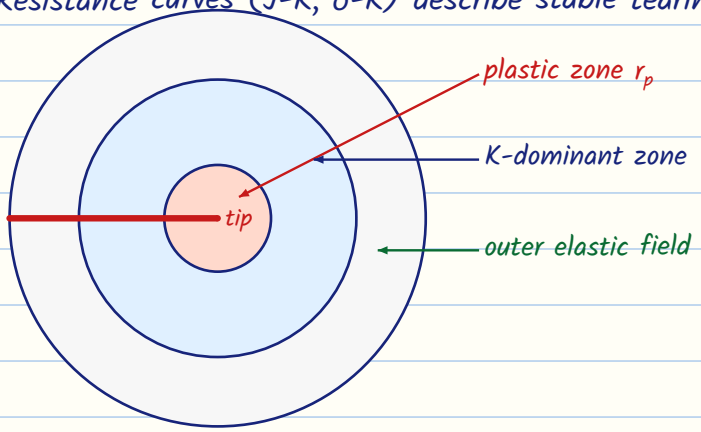


Fig 1.4 Crack-tip zones : K-dominant region vs plastic zone

Feature	LEFM	EPFM
Material behaviour	Linear elastic	Elastic-plastic / non-linear elastic
Plastic zone	Small (SSY), r_p << a	Large, comparable to a, ligament
Parameters	K, G	J, CTOD (δ), tearing modulus T
Fracture criterion	$K = K_{Ic}$, $G = G_c$	$J = J_{Ic}$, $\delta = \delta_c$
Typical materials	High-strength steel, Al, glass, ceramics	Mild steel, stainless steel, Ti, polymers
Standards	ASTM E399	ASTM E1820, BS 7448
Specimen size	Large (plane strain)	Smaller specimens allowed
Mathematics	Simple, closed-form	Needs numerical / FEM, handbooks

Remember

> LEFM ⊂ EPFM : when plasticity → 0, $J \to G = K^2/E'$ ($E' = E$ plane stress; $E/(1-\nu^2)$ plane strain).

> Unit of K : MPa√m (= 10^6 N $m^{-3/2}$); unit of G : N/m (J/m^2); J : N/m; δ : m.

5. Fatigue Crack Growth Model

Fatigue = progressive, localised damage and failure under repeated (cyclic) stress which may be well below the yield strength. About 80-90 % of service failures are fatigue related.

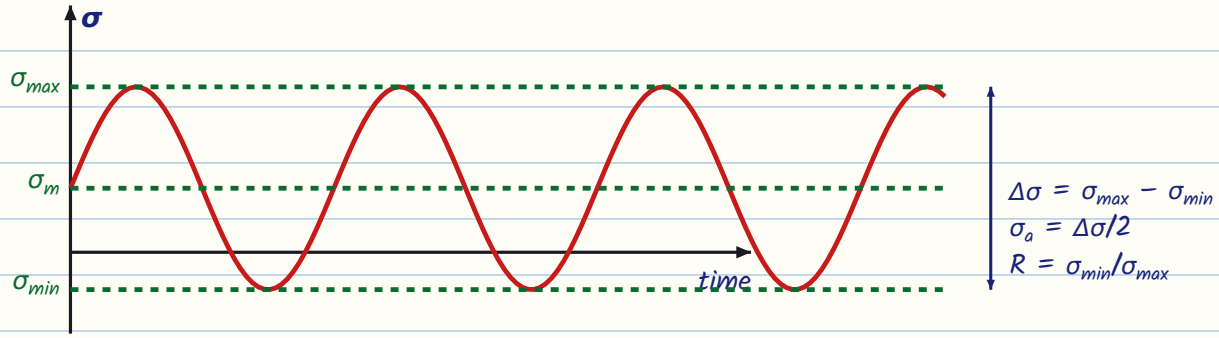


Fig 1.5 Constant-amplitude stress cycle and its parameters

$$\Delta\sigma = \sigma_{max} - \sigma_{min} \quad \sigma_a = \Delta\sigma/2 \quad \sigma_m = (\sigma_{max}+\sigma_{min})/2 \quad R = \sigma_{min}/\sigma_{max}$$

5.1 Three stages of fatigue life

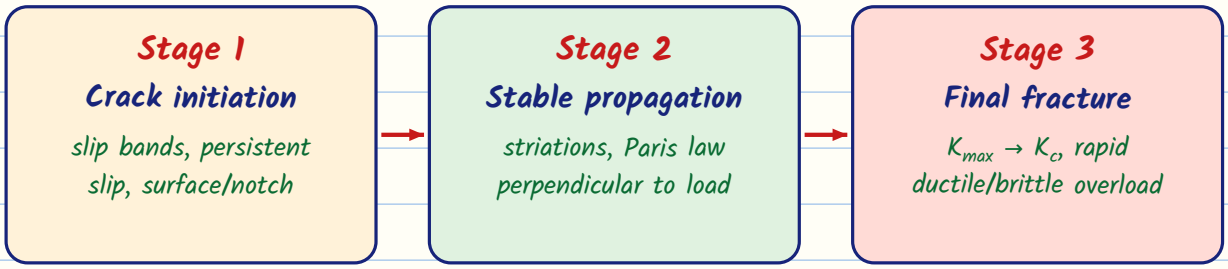


Fig 1.6 Stages of fatigue failure

- **Stage I** (initiation): slip bands, intrusions/extrusions at surface; ~ 90 % of life in high-cycle fatigue; initiates at notches, inclusions.
- **Stage II** (propagation): crack grows ⊥ to max tensile stress; leaves **striations** (one per cycle) and **beach marks** (load changes); governed by ΔK.
- **Stage III** (final fracture): when K_{max} reaches K_c (or net-section yields) – sudden fracture, ductile or brittle.

5.2 S-N approach vs fracture mechanics approach

S-N / strain-life	Fracture mechanics (damage tolerance)
Total life to failure of smooth specimen (initiation dominated)	Crack propagation life from an initial flaw a_0 to a_c
Uses stress amplitude, endurance limit, Goodman/Miner	Uses ΔK, da/dN, Paris law; inspection-based
No crack assumed	Crack assumed (or detected by NDT)

5.3 Fatigue crack growth model (a vs N)

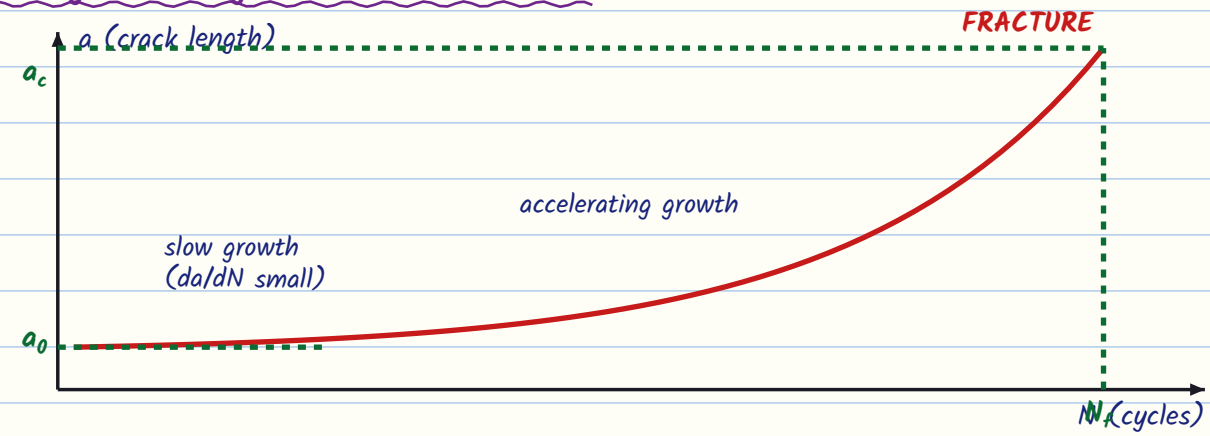


Fig 1.7 Crack length vs number of cycles

Experiment: pre-cracked specimen under constant $\Delta\sigma$; measure a vs N. Slope of the curve = **crack growth rate da/dN**. The rate increases as a grows because $\Delta K = \Delta\sigma\, Y \sqrt{(\pi a)}$ grows with a.

Fundamental hypothesis: the crack-growth rate is controlled by the stress-intensity range:

$$da/dN = f(\Delta K, R) \text{ where } \Delta K = K_{max} - K_{min} = \Delta\sigma \cdot Y \cdot \sqrt{(\pi a)}$$

For $R = 0$, $K_{min} = 0$ so $\Delta K = K_{max}$. (Only the tensile part of the cycle is counted; compressive part ignored.)

5.4 Paris-Erdogan law (basic crack-growth model)

$$da/dN = C\,(\Delta K)^m$$ Paris & Erdogan (1963)

- C, m = material constants (m ≈ 2-4 for metals, ~ 3 for steels; ceramics/polymers m much higher). Units of C depend on m; for da/dN in m/cycle & ΔK in MPa√m.
- Valid in the stable (Region II) growth range only; details & sigmoidal curve in Unit 5.

5.5 Fatigue life by integration

Separate variables and integrate from initial a_0 to critical a_c:

$$N_f = \frac{\int da \;(a_0 \to a_c)}{C\,(\Delta\sigma\, Y \sqrt{(\pi a)})^m}$$

If Y is constant and $m \neq 2$:

$$N_f = \frac{a_c^{\,1-m/2} - a_0^{\,1-m/2}}{C\,(Y\,\Delta\sigma\,\sqrt{\pi})^m\,(1 - m/2)} \qquad m \neq 2$$

For $m = 2$: $N_f = [1 / (C\,\pi\,Y^2\,\Delta\sigma^2)] \cdot \ln(a_c / a_0)$.

Critical crack length from fracture criterion $K_{max} = K_c$:

$$a_c = \frac{1}{\pi}\left(K_c / (Y\,\sigma_{max})\right)^2$$

Steps for a fatigue-life problem (EXAM)

> 1) Get Y (geometry), σ_{max}, $\Delta\sigma$, K_c, C, m, a_0 from data.
> 2) Compute $a_c = (1/\pi)(K_c/(Y\sigma_{max}))^2$.
> 3) Check whether Y varies with a; if constant use closed form, else integrate numerically (stepwise).
> 4) N_f from the formula above; apply safety factor (e.g. $N_f/2$ or /4) to fix inspection interval.

Worked Example 1

A wide steel plate (Y = 1.0) has an initial central crack $2a_0$ = 4 mm and is loaded by $\Delta\sigma$ = 100 MPa (R = 0, σ_{max} = 100 MPa). K_c = 80 MPa√m; C = 6.9×10^{-12} , m = 3 (m/cycle, MPa√m). Find a_c and N_f.

Sol. a_0 = 2 mm = 0.002 m.

$a_c = (1/\pi)(80/100)^2 = 0.64/\pi =$ **0.2037 m** (≈ 204 mm).

$1 - m/2 = -0.5$. $a_0^{-0.5} = 22.36$; $a_c^{-0.5} = 2.215$.

Denominator: $C\,(\Delta\sigma\sqrt{\pi})^3\,(-0.5) = 6.9\times10^{-12} \times (100\times1.7725)^3 \times (-0.5) = 6.9\times10^{-12} \times 5.569\times10^{6} \times (-0.5) = -0.01921$.

Numerator = 2.215 − 22.36 = −20.15.

$N_f = (-20.15)/(-0.01921) \approx 1.05 \times 10^6$ cycles. (Most life is spent while the crack is small!)

Worked Example 2 (critical size)

An edge crack in a large plate (Y = 1.12) carries σ = 250 MPa; K_{Ic} = 55 MPa√m. Critical crack size?

$a_c = (1/\pi)(K_{Ic}/(Y\sigma))^2 = (1/\pi)(55/(1.12\times250))^2 = (1/\pi)(0.1964)^2 =$ **0.01228 m ≈ 12.3 mm**.

Worked Example 3 (edge crack, inspection interval)

Large plate, single edge crack, Y = 1.12, $\Delta\sigma = \sigma_{max}$ = 150 MPa (R = 0). K_c = 60 MPa√m, C = 1×10^{-11}, m = 3. Initial crack a_0 = 1 mm (largest flaw missed by NDT). Find N_f and a safe inspection interval.

$a_c = (1/\pi)(60/(1.12\times150))^2 = (1/\pi)(0.3571)^2 =$ **0.0406 m = 40.6 mm**.

$Y\,\Delta\sigma\,\sqrt{\pi} = 1.12\times150\times1.7725 = 297.8$; $(297.8)^3 = 2.641\times10^7$.

$N_f = (a_c^{-0.5} - a_0^{-0.5}) / [C\,(Y\Delta\sigma\sqrt{\pi})^3\,(-0.5)] = (4.963 - 31.62)/(1\times10^{-11} \times 2.641\times10^7 \times (-0.5)) = (-26.66)/(-1.32\times10^{-4}) \approx$ **2.0×10^5 cycles**.

With a scatter factor of 2 ($N_f/2$) inspect every ≈ 1.0×10^5 cycles. If NDT can find 5 mm

cracks, recompute with a_0 = 5 mm for a more realistic (and longer) schedule.

Worked Example 4 (step-by-step / numerical integration idea)

When Y depends on a, divide the crack range into steps, compute ΔK at the mean a of each step and use ΔN = Δa / (C ΔK^m).

a (mm)	ΔK = Δσ√(πa) (MPa√m)	da/dN = C ΔK^3 (m/cycle)
2	7.93	3.44×10^{-9}
4	11.21	9.72×10^{-9}
8	15.85	2.75×10^{-8}
16	22.42	7.78×10^{-8}
32	31.71	2.20×10^{-7}

(Δσ = 100 MPa, Y = 1, $C = 6.9\times10^{-12}$, m = 3.) Doubling a raises da/dN by $2^{1.5} = 2.83\times$. Cycles to grow 2→4 mm ≈ 3.4×10^{5}; 4→8 mm ≈ 2.4×10^{5}: crack growth **accelerates**.

SA. More on LEFM / EPFM and Fatigue

A. Plane stress and plane strain

- **Thin plate** : free to contract through thickness → $\sigma_z = 0$ → **plane stress**; large plastic zone, shear lips, higher toughness K_c.
- **Thick plate** : through-thickness contraction constrained → $\varepsilon_z = 0$, $\sigma_z = \nu(\sigma_x + \sigma_y)$ → **plane strain**; high triaxiality, small plastic zone, flat fracture, lowest toughness K_{Ic} (a conservative material property).
- Thickness for valid K_{Ic} (ASTM E399): $B, a, (W - a) \geq 2.5\,(K_{Ic} / \sigma_{ys})^2$.
- Plastic-zone size (Irwin): plane stress $r_p = (1/2\pi)(K/\sigma_{ys})^2$; plane strain ≈ one-third of it.

B. Introduction to CTOD and J-integral (EPFM parameters)

- **CTOD (δ)** (Wells, 1961): crack faces blunt and separate before fracture; opening at the original tip = δ. Fracture when $\delta = \delta_c$. In SSY: $\delta = K^2 / (\sigma_{ys} E')$ ($\approx G / \sigma_{ys}$).
- **J-integral** (Rice, 1968): path-independent contour integral = energy-release rate for non-linear elastic body. J = G in linear elasticity. Fracture when $J = J_{Ic}$.
- Experimentally J_{Ic} is measured from a J-R curve (unloading compliance); $K_{Jc} = \sqrt{(E' J_c)}$.

C. Factors affecting fatigue crack growth

Factor	Effect
Stress ratio R / mean stress	Higher R → higher da/dN (K_{max} larger, less closure); threshold ΔK_{th} falls
Overload (tensile)	Plastic zone causes compressive residual stress → crack **retardation** or arrest

Environment	Corrosion fatigue, hydrogen: da/dN increases, lower ΔK_{th}
Frequency & temperature	Low f / high T → time-dependent (creep, oxidation) increases growth
Microstructure, grain size	Coarse grains → better threshold; strength has weak effect in Region II
Thickness	Thin plates (plane stress) show more closure → slower growth

D. Fatigue fracture surface features

- **Beach (clamshell) marks** – macroscopic bands from changes in load or stops/starts; show crack origin and direction (arcs centred on origin).
- **Striations** – microscopic ripples, **one per load cycle** (seen under SEM/TEM). Spacing ≈ da/dN, so striation spacing can be used to estimate growth rate in failure analysis.
- **Ratchet marks** – multiple initiation sites at a notch or corner.
- **Final fast-fracture zone** – rough ductile/brittle region; its size shows how highly the part was stressed.

Cost of fracture

> A US study (Battelle/NBS, 1983) estimated fracture costs ≈ 4 % of GNP (~ $119 billion per year); about a third could be avoided with present knowledge.

> Fracture-control plans (design → material selection → NDT → inspection → repair) are therefore mandatory in aerospace, nuclear, pipelines & offshore.

SB. Glossary & 2-mark Answers

Term	Meaning
Fracture toughness	Resistance of a cracked material to crack extension (K_{Ic}, G_c, J_{Ic})
Stress intensity factor K	Parameter measuring the severity of the crack-tip stress field; $K = \sigma Y\sqrt{(\pi a)}$
Energy release rate G	Energy released per unit new crack area; $G = \pi\sigma^2 a/E$ (plane stress)
Plastic zone	Region ahead of tip where σ exceeds yield stress
Damage tolerance	Ability of a structure to carry load safely with cracks until detected/repaired
ΔK_{th}	Threshold stress-intensity range below which cracks do not grow
Striation	Fatigue marking left by one load cycle
Residual strength	Remaining strength of a cracked structure as a function of crack size

Q: Why do cracks usually start at the surface? Maximum stresses (bending/torsion),

stress raisers, free-surface slip, environment attack.

Q: Is a compressive stress cycle harmful? Crack faces close and carry compression, so only the tensile part contributes to ΔK.

SC. Model Answers (10-mark style)

Q. Explain the fracture-mechanics approach to design.

1. Assume a flaw of size a exists (largest one NDT can miss or has detected).
2. Compute $K = \sigma Y \sqrt{(\pi a)}$ at the critical location for the design load (Y from handbook).
3. Compare with K_{Ic} (plane strain) or K_c. Require $K \leq K_{Ic}$ / FoS.
4. If cyclic loading, compute ΔK and use $da/dN = C \Delta K^m$ to find the number of cycles to reach a_c.
5. Fix inspection interval ≤ N_f / safety factor; repair or replace when a crack is found above $a_{detectable}$.

Residual strength diagram

> Plot of failure stress $\sigma_f = K_c / (Y\sqrt{(\pi a)})$ against crack size a. It falls as a increases: large cracks → low residual strength.

> At $a \to 0$ the curve is limited by σ_{ys} (net-section yield), so the true curve is cut off by a horizontal line at yield/flow stress.

> Fail-safe design keeps the operating stress below the curve for the largest crack that can exist between inspections.

Q. Write notes on Liberty ship failures.

- Welded all-steel cargo ships built in WWII at high speed; all-welded hull let a crack run across several plates (riveted ships arrest cracks at plate edges).
- Steel (low Mn:C ratio, high S) had a high transition temperature, near sea-water temperature (~ 0-5 °C in the North Atlantic) → low toughness.
- Stress raisers: square hatch corners, bad weld details & residual welding stresses.
- Remedies: better steel (Mn:C ≥ 2.5), rounded hatch corners, riveted crack arrester strakes, welding-procedure control, Charpy requirement at service temperature. This work (Irwin, Wells, Naval Research Lab) was the birth of modern fracture mechanics.

Q. Why is K_{Ic} a conservative property?

Plane-strain gives the highest constraint and lowest fracture toughness; thinner sections have $K_c > K_{Ic}$. Using K_{Ic} in design is therefore safe (conservative) for all thicknesses.

Quick MCQs

1) Unit of K_I : (a) MPa (b) **MPa√m** (c) N/m (d) J. Ans: b

2) Mode of loading where crack faces separate normal to the plane : **Mode I**.

3) The Paris law is applicable to : **Region II (stable growth)** of the da/dN–ΔK curve.

4) The crack-growth rate is a function of : **ΔK (and R)**, not of Δσ alone.

5) BCC metals exhibit DBTT; FCC metals (Al, Cu, austenitic steel) generally do **not**.

6. Important Exam Questions (Unit I)

1. Explain with examples three spectacular failures and the lessons learnt for fracture-control design.
2. Differentiate between LEFM and EPFM. State the parameters and applicability of each.
3. Why is the conventional stress-based design inadequate? Explain the three variables of fracture mechanics approach.
4. Describe the three modes of fracture with sketches.
5. Explain the stages of fatigue failure and derive an expression for fatigue life from the Paris law.
6. Compare safe-life, fail-safe and damage-tolerant design philosophies.
7. Explain ductile-brittle transition temperature and the factors that influence it.

Quick revision: $K = \sigma Y\sqrt{(\pi a)}$; $K = K_c \rightarrow$ fracture ; $da/dN = C\ \Delta K^m$; $N_f = (a_c^{1-m/2} - a_0^{1-m/2}) / (C\ (Y\Delta\sigma\sqrt{\pi})^m\ (1 - m/2))$.